

Polar Semirings*

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Abstract

We introduce *Polar Semirings*, a class of algebraic structures in which polarity—an order-reversing involution—is taken as a primitive operation rather than derived from residuation. A Polar Semiring is axiomatized by eleven equational and quasi-equational axioms over the primitive signature $(X, \vee, \otimes, *, e_\vee, e_\otimes)$; the meet, the plus operation, and their units are obtained by conjugation under the polarity. We show that the axiom system is independent, exhibit canonical models, and prove a Fundamental Structure Theorem: every Polar Semiring canonically induces a dual Polar Semiring, with the polarity acting as an isomorphism and the two induced orders coinciding. We then position the class within the landscape of involutive algebraic structures: every bounded Girard algebra induces a Polar Semiring, whereas there exists a Polar Semiring on the unit interval that cannot be extended to a bounded Girard algebra. Consequently, bounded Girard algebras form a proper subclass, and Polar Semirings constitute a genuine residuation-free generalization.

[AI-check: Add MSC2020 codes (suggest: 06B05, 06D30, 16Y60, 03G10) and keywords (polarity; involution; semiring; Girard algebra; residuation) once the target journal is chosen.]

1 Introduction

Order-reversing involutions pervade algebra. In lattice theory they appear as the involutions of De Morgan algebras and of lattices with involution [5, 7]; in the theory of residuated structures they arise as the negations of involutive residuated lattices and Girard-style algebras [1, 3, 8]. In the residuated tradition, however, the involution is not a primitive operation: it is manufactured from the residual, by fixing a distinguished element $e_\oplus$ and setting $a^* := a \rightarrow e_\oplus$. The involution thus inherits its entire behaviour—antitonicity, the De Morgan laws, the duality between the multiplicative operations—from the residuation law, and it is natural to ask how much of this behaviour genuinely depends on residuation.

The answer proposed in this paper is: very little. Following Birkhoff [2], we call a *polarity* an involution that is compatible with an ambient order, in the sense that $a \leq b$ if and only if $b^* \leq a^*$. Our thesis is that polarity is a primitive algebraic notion in its own right, and that the familiar dual architecture generated by a negation—dual monoid, dual semiring, De Morgan laws, order reversal—can be reconstructed from a small number of axioms that never mention a residual.

To substantiate this thesis we introduce *Polar Semirings*: idempotent commutative semirings equipped with a primitive polarity. A Polar Semiring is specified by eleven axioms over the signature $(X, \vee, \otimes, *, e_\vee, e_\otimes)$, of which eight are the standard monoid and semiring identities of the underlying operations, and only three—the polarity axiom and the two distributivity–absorption axioms linking $\otimes$ with $\vee$ —carry genuine compatibility content. The meet, the plus operation, and their units are not primitive: they are defined by conjugation under the polarity, and their entire algebra is derived.

*[AI-check: Subtitle intentionally left open per final-rules §0; add one if the journal requires it.]

Contributions. The paper makes three contributions.

1. *Definition and minimality* (Section 2). We axiomatize Polar Semirings by eleven axioms, exhibit canonical models, and prove that the axiom system is independent: no axiom is a consequence of the remaining ten.
2. *Architecture* (Section 3). We show that the axioms decompose into three layers: an algebraic layer (the polarity-free semiring reduct), a reflection layer, in which the involution axiom alone generates the dual operations and all De Morgan laws at no axiomatic cost, and an integration layer, in which the single polarity axiom fuses the reflected order with the original one. The decomposition culminates in the Fundamental Structure Theorem: every Polar Semiring canonically induces a dual Polar Semiring, the polarity is an isomorphism onto the dual, and the two induced orders coincide.
3. *Positioning* (Section 4). We locate Polar Semirings relative to bounded Girard algebras. Every bounded Girard algebra, upon forgetting its residual, satisfies all eleven axioms; and there exists a Polar Semiring on the unit interval that cannot be extended to a bounded Girard algebra. Hence bounded Girard algebras form a proper subclass, and Polar Semirings are a strict, residuation-free generalization.

Related structures. The order-theoretic reduct of a Polar Semiring is a bounded lattice with an antitone involution satisfying the De Morgan laws, that is, a bounded i-lattice in the sense of Kalman [5]; distributivity is not assumed, so De Morgan algebras [7] and Boolean algebras arise from Polar Semirings only after adding further axioms. On the multiplicative side, semirings with involution [4] retain the involution but drop the order-compatibility axiom entirely. A brief orientation table comparing these classes precedes the detailed comparison with bounded Girard algebras in Section 4. Throughout the paper we maintain a strict terminological distinction, elaborated in Section 3: a structure is called *involutive* when it satisfies only the equation $(a^*)^* = a$, and *polar* when the involution is additionally compatible with the induced order, in Birkhoff’s classical sense [2].

Outline. Section 2 defines Polar Semirings, presents basic models, and proves independence of the axioms. Section 3 develops the three-layer architectural analysis and proves the Fundamental Structure Theorem. Section 4 carries out the comparison with bounded Girard algebras, proving both the inclusion and its strictness. Section 5 summarizes and discusses future directions.

2 Polar Semirings

This section introduces the central object of the paper. We first state the eleven axioms (Section 2.1), then show that they are non-empty by exhibiting canonical models (Section 2.2), and finally prove that the axiom system is independent (Section 2.3).

2.1 Definition

Definition 2.1 (Polar Semiring). A *Polar Semiring* is a structure

$$(X, \vee, \otimes, *, e_\vee, e_\otimes),$$

where $\vee$ and $\otimes$ are binary operations, $*$ is a unary operation, and $e_\vee, e_\otimes \in X$, satisfying the following eleven axioms for all $a, b, c \in X$:

$$\begin{aligned}
a \vee a &= a, & (J0) \\
(a \vee b) \vee c &= a \vee (b \vee c), & (J1) \\
a \vee b &= b \vee a, & (J2) \\
a \vee e_\vee &= a, & (J3) \\
(a \otimes b) \otimes c &= a \otimes (b \otimes c), & (T1) \\
a \otimes b &= b \otimes a, & (T2) \\
a \otimes e_\otimes &= a, & (T3) \\
a \otimes (b \vee c) &= (a \otimes b) \vee (a \otimes c), & (TJ1) \\
a \otimes e_\vee &= e_\vee, & (TJ2) \\
(a^*)^* &= a, & (P) \\
a \vee b = b &\iff b^* \vee a^* = a^*. & (PJ)
\end{aligned}$$

The operation $*$ is called the *polarity* of the structure. We abbreviate “Polar Semiring” by PS.

Axioms (J0)–(J3) state that $(X, \vee, e_\vee)$ is an idempotent commutative monoid; (T1)–(T3) state that $(X, \otimes, e_\otimes)$ is a commutative monoid; (TJ1)–(TJ2) link the two, so that the polarity-free reduct $(X, \vee, \otimes, e_\vee, e_\otimes)$ is an idempotent commutative semiring [4]. Axiom (P) makes $*$ an involution, and (PJ) is the only axiom relating the polarity to $\vee$; its meaning becomes transparent once the induced order is introduced.

Definition 2.2 (Derived operations and orders). Let $(X, \vee, \otimes, *, e_\vee, e_\otimes)$ be a Polar Semiring. Define the *meet*, the *plus*, and the derived units by

$$a \wedge b := (a^* \vee b^*)^*, \quad a \oplus b := (a^* \otimes b^*)^*, \quad e_\wedge := e_\vee^*, \quad e_\oplus := e_\otimes^*, \quad (1)$$

and the *join order* and *meet order* by

$$a \leq b :\iff a \vee b = b, \quad a \leq^* b :\iff a \wedge b = a. \quad (2)$$

With (2), axiom (PJ) reads: $a \leq b$ if and only if $b^* \leq a^*$, that is, $*$ is an antitone involution on $(X, \leq)$. This is precisely the classical notion of a polarity on an ordered set [2], which motivates our terminology. We emphasize that $\wedge, \oplus, e_\wedge$, and $e_\oplus$ are not part of the signature: all of their properties must be, and in Section 3 will be, derived from the eleven axioms. Note also that the axioms do not force $e_\vee, e_\wedge, e_\otimes, e_\oplus$ to be pairwise distinct; in Example 2.5 below one has $e_\otimes = e_\oplus$.

2.2 Basic models

The following lemma produces a large supply of models on chains.

Lemma 2.3. *Let $(X, \leq)$ be a totally ordered set with least element $e_\vee$, let $\vee := \max$, and let $(X, \otimes, e_\otimes)$ be a commutative monoid such that $\otimes$ is monotone in each argument and $a \otimes e_\vee = e_\vee$ for all a . If $*$: $X \rightarrow X$ is an order-reversing involution, then $(X, \vee, \otimes, *, e_\vee, e_\otimes)$ is a Polar Semiring.*

Proof. Axioms (J0)–(J3) hold for $\max$ with least element $e_\vee$, and (T1)–(T3), (TJ2), (P) hold by hypothesis. For (TJ1), monotonicity of $\otimes$ gives $a \otimes \max(b, c) = \max(a \otimes b, a \otimes c)$ on a chain, since $\max(b, c) \in \{b, c\}$. Finally, since $*$ is an order-reversing bijection, $a \leq b$ holds if and only if $b^* \leq a^*$, which is (PJ) by (2). $\square$

Example 2.4 (Boolean model). Take $X = \{0, 1\}$ with $\vee = \max$, $\otimes = \min$, $x^* = 1 - x$, $e_\vee = 0$, $e_\otimes = 1$. By Lemma 2.3 this is a Polar Semiring; the derived operations satisfy $\wedge = \otimes = \min$ and $\oplus = \vee = \max$, recovering the two-element Boolean algebra.

Example 2.5 (Tropical model). Take $X = \mathbb{R} \cup \{-\infty, +\infty\}$ with $\vee = \max$, $a \otimes b = a + b$, $x^* = -x$, $e_\vee = -\infty$, $e_\otimes = 0$, under the convention $(-\infty) + x = -\infty$ for every x , including $x = +\infty$. Lemma 2.3 applies, so this is a Polar Semiring with $e_\wedge = +\infty$ and $e_\oplus = 0^* = 0 = e_\otimes$. The derived plus is $a \oplus b = a + b$ under the opposite convention $(+\infty) + x = +\infty$. [AI-check: The convention $(-\infty) + (+\infty) = -\infty$ for $\otimes$ (and dually $+\infty$ for $\oplus$) is forced by (TJ2); please confirm you are happy stating it explicitly here rather than in a footnote.]

Example 2.6 (Quadratic model on the unit interval). Take $X = [0, 1]$ with $\vee = \max$, $a \otimes b = ab$ (ordinary multiplication), and

$$x^* := (1 - \sqrt{x})^2, \quad (3)$$

with $e_\vee = 0$ and $e_\otimes = 1$. The map (3) is strictly decreasing on $[0, 1]$ with $0^* = 1$ and $1^* = 0$, and it is an involution: since $1 - \sqrt{x} \geq 0$ on $[0, 1]$,

$$(x^*)^* = \left(1 - \sqrt{(1 - \sqrt{x})^2}\right)^2 = (1 - (1 - \sqrt{x}))^2 = x.$$

Multiplication is monotone and absorbs 0, so Lemma 2.3 applies. Here $e_\wedge = 1$ and $e_\oplus = 0$. This model plays a decisive role in Section 4.3.

2.3 Independence

Theorem 2.7 (Independence). *The axiom system of Definition 2.1 is independent: for each of the eleven axioms there exists a finite structure $(X, \vee, \otimes, *, e_\vee, e_\otimes)$ satisfying the remaining ten axioms but not the selected one.*

Proof. For each axiom, a finite countermodel was obtained by an exhaustive search with the model finder Mace4, and the non-derivability claims were cross-checked with the theorem prover Prover9 [6]. The countermodels have at most [AI-check: fill in the maximal cardinality] elements and are listed in [AI-check: decide: appendix with the eleven model tables, or supplementary material]. [AI-check: Per final-rules §6.2, the original Mace4/Prover9 runs used the obsolete axiom names (M0–M3, S1–S3, SM1–SM2, PM) and the obsolete choice of primitives ($\wedge$ primitive). The runs must be re-executed in the present signature (primitives $\vee, \otimes, *, e_\vee, e_\otimes$; axiom names J0–J3, T1–T3, TJ1–TJ2, P, PJ) before this theorem is submitted.] $\square$

Remark 2.8. Among the eleven axioms, exactly three carry compatibility content between distinct primitives: (PJ), linking $*$ with $\vee$, and (TJ1)–(TJ2), linking $\otimes$ with $\vee$. The remaining eight are the internal monoid axioms of the two layers together with the involution equation (P). This observation organizes the whole of Section 3 and will be echoed in Section 4.

We close the section by announcing the main structural result, whose proof occupies Section 3.

Theorem 2.9 (Fundamental Structure Theorem; proved as Theorem 3.6). *Every Polar Semiring $(X, \vee, \otimes, *, e_\vee, e_\otimes)$ canonically induces a dual Polar Semiring $(X, \wedge, \oplus, *, e_\wedge, e_\oplus)$; the polarity is an isomorphism between the two structures, and the meet order $\leq^*$ coincides with the join order $\leq$.*

3 Architectural Analysis

The purpose of this section is to explain why the eleven axioms of Definition 2.1 decompose into a modular architecture, and to prove the Fundamental Structure Theorem announced in Theorem 2.9. The decomposition has three layers. The *algebraic layer* (Section 3.1) is the polarity-free reduct: two monoid modules and the two axioms linking them. The *reflection layer* (Section 3.2) shows that the involution axiom (P) alone reflects each module into a dual module and yields every De Morgan law at no further axiomatic cost. The *integration layer* (Section 3.3) shows that the single axiom (PJ) integrates the reflected order with the original one, producing a lattice and completing the structure. Throughout, recall from Remark 2.8 that only (PJ), (TJ1), and (TJ2) carry compatibility content; the architecture below makes this precise.

3.1 Algebraic layer

The algebraic layer consists of two independent modules and their interaction:

1. the *join module* $(X, \vee, e_\vee)$, an idempotent commutative monoid by (J0)–(J3);
2. the *tensor module* $(X, \otimes, e_\otimes)$, a commutative monoid by (T1)–(T3);
3. the interaction axioms (TJ1)–(TJ2), which make $(X, \vee, \otimes, e_\vee, e_\otimes)$ an idempotent commutative semiring [4].

The join module already carries order-theoretic information: by (J0)–(J2), the relation $\leq$ of (2) is a partial order (reflexivity is (J0), antisymmetry follows from (J2), transitivity from (J1)), with least element $e_\vee$ by (J3), and $a \vee b$ is the least upper bound of a and b . The interaction axioms give monotonicity of the tensor:

Lemma 3.1. *In any Polar Semiring, $a \leq b$ implies $a \otimes c \leq b \otimes c$.*

Proof. If $a \vee b = b$, then by (TJ1) and (T2), $(a \otimes c) \vee (b \otimes c) = (a \vee b) \otimes c = b \otimes c$. $\square$

Nothing in this layer mentions $*$: it is ordinary semiring theory.

3.2 Reflection layer

We now add the involution axiom (P) and nothing else, and show that it *reflects* each module of the algebraic layer into a dual module, with all duality laws obtained for free. We first fix terminology. Following the historical order of the notions—involutions belong to pure algebra, polarities to order theory [2]—we call a structure *involutive* when it satisfies only $(a^*)^* = a$, and reserve the adjective *polar* for structures in which the involution is additionally compatible with the induced order via (PJ). Thus an *involutive commutative monoid* is a commutative monoid equipped with an involution; no order is involved, and indeed at this level no order need exist.

Proposition 3.2 (Duality laws). *Let $(X, \vee, e_\vee)$ and $(X, \otimes, e_\otimes)$ be commutative monoids and let $*$ satisfy (P). With the derived operations (1), the following hold for all $a, b \in X$:*

$$(a \wedge b)^* = a^* \vee b^*, \tag{4}$$

$$(a \oplus b)^* = a^* \otimes b^*, \tag{5}$$

$$(a \vee b)^* = a^* \wedge b^*, \tag{6}$$

$$(a \otimes b)^* = a^* \oplus b^*, \tag{7}$$

$$e_\vee^* = e_\wedge, \quad e_\wedge^* = e_\vee, \quad e_\otimes^* = e_\oplus, \quad e_\oplus^* = e_\otimes. \tag{8}$$

Proof. For (4), apply (P) to the definition: $(a \wedge b)^* = ((a^* \vee b^*)^*)^* = a^* \vee b^*$. For (6), compute $a^* \wedge b^* = ((a^*)^* \vee (b^*)^*)^* = (a \vee b)^*$ by (P). Equations (5) and (7) are identical computations with $\otimes, \oplus$ in place of $\vee, \wedge$. In (8), the first and third equations are the definitions in (1), and the other two follow by (P). $\square$

Proposition 3.3 (Reflected modules). *Under the hypotheses of Proposition 3.2:*

1. $(X, \wedge, e_\wedge)$ is a commutative monoid, the meet module;
2. $(X, \oplus, e_\oplus)$ is a commutative monoid, the plus module;
3. if moreover (J0) holds, then $\wedge$ is idempotent.

Proof. Commutativity of $\wedge$ is immediate from (J2). For associativity, using (4),

$$(a \wedge b) \wedge c = ((a \wedge b)^* \vee c^*)^* = ((a^* \vee b^*) \vee c^*)^*,$$

and (J1) lets the inner join reassociate; reversing the steps gives $a \wedge (b \wedge c)$. For the unit, $a \wedge e_\wedge = (a^* \vee e_\wedge^*)^* = (a^* \vee e_\vee^*)^* = (a^*)^* = a$ by (8), (J3), and (P). Item (2) is the same argument in the tensor module, using (T1)–(T3). For (3), $a \wedge a = (a^* \vee a^*)^* = (a^*)^* = a$ by (J0) and (P). $\square$

Two comments are in order. First, everything above is a consequence of the involution equation alone; the reflection layer is *axiomatically free*. Second, at this stage the two relations $\leq$ and $\leq^*$ of (2) are both available, but nothing yet guarantees that they coincide, nor that $\wedge$ computes greatest lower bounds with respect to $\leq$. That is precisely the task of the third layer.

3.3 Integration layer

The final axiom (PJ) is the first, and only, axiom relating the polarity to the join module, and it is not derivable from the previous layers (Theorem 2.7). Its effect is to integrate the reflected order with the original one.

Proposition 3.4 (Order compatibility). *In the presence of the other ten axioms, the following are equivalent:*

1. (PJ): $a \vee b = b \iff b^* \vee a^* = a^*$;
2. $a \vee b = b \iff b \wedge a = a$;
3. $a \leq b \iff b^* \leq^* a^*$;
4. the absorption laws: $a \wedge (a \vee b) = a$ and $a \vee (a \wedge b) = a$.

Proof. (1) $\Leftrightarrow$ (2): by (1) and (P), $b \wedge a = a$ holds if and only if $(b^* \vee a^*)^* = a$, if and only if $b^* \vee a^* = a^*$; so (2) is a literal restatement of (1).

(1) $\Leftrightarrow$ (3): unfolding (2), $b^* \leq^* a^*$ means $b^* \wedge a^* = b^*$, that is, $(b \vee a)^* = b^*$ by (6), that is, $b \vee a = a$ by injectivity of $*$; renaming variables shows (3) to be a restatement of (1) as well.

(2) $\Rightarrow$ (4): since $a \leq a \vee b$ by (J0)–(J1), item (2) yields $(a \vee b) \wedge a = a$, which is the first absorption law by commutativity of $\wedge$. For the second, (2) applied in the direction “ $b \wedge a = a \Rightarrow a \vee b = b$ ” with the pair $(a \wedge b, a)$: indeed $a \wedge (a \wedge b) = a \wedge b$ by idempotence and associativity of $\wedge$ (Proposition 3.3), so $(a \wedge b) \vee a = a$, which is the second absorption law.

(4) $\Rightarrow$ (2): if $a \vee b = b$, then $a \wedge b = a \wedge (a \vee b) = a$, so $b \wedge a = a$. Conversely, if $b \wedge a = a$, then $a \vee b = (b \wedge a) \vee b = b$ by the second absorption law and commutativity. $\square$

Proposition 3.5 (Order structure). *In any Polar Semiring:*

1. $\leq$ is a partial order and $\leq = \leq^*$;
2. $e_\vee \leq a \leq e_\wedge$ for all a ;
3. $a \leq b$ implies $a \wedge c \leq b \wedge c$ and $a \oplus c \leq b \oplus c$;
4. $a \vee b$ is the least upper bound and $a \wedge b$ the greatest lower bound of $\{a, b\}$ with respect to $\leq$; consequently $(X, \vee, \wedge)$ is a (possibly non-distributive) lattice.

Proof. (1) That $\leq$ is a partial order was noted in Section 3.1. By Proposition 3.4(2), $a \leq b$ if and only if $a \wedge b = a$, which is $a \leq^* b$; hence the two orders coincide.

(2) $e_\vee \leq a$ is (J3) with (J2). For $a \leq e_\wedge$: by (PJ) this is equivalent to $e_\wedge^* \vee a^* = a^*$, that is, $e_\vee \vee a^* = a^*$, which holds by (J3).

(3) Suppose $a \leq b$, so $a \wedge b = a$ by Proposition 3.4(2). Then $(a \wedge c) \wedge (b \wedge c) = (a \wedge b) \wedge (c \wedge c) = a \wedge c$ by Proposition 3.3, whence $a \wedge c \leq^* b \wedge c$, and we conclude by (1). For $\oplus$: from $a \leq b$, (PJ) gives $b^* \leq a^*$, then Lemma 3.1 gives $b^* \otimes c^* \leq a^* \otimes c^*$, and applying (PJ) once more together with (5) yields $a \oplus c \leq b \oplus c$.

(4) The least-upper-bound property of $\vee$ is standard in idempotent commutative monoids: $a, b \leq a \vee b$, and if $a, b \leq c$ then $(a \vee b) \vee c = c$. For $\wedge$, dualize: by (PJ) and (6), lower bounds of $\{a, b\}$ correspond under $*$ to upper bounds of $\{a^*, b^*\}$, and the correspondence sends $a \wedge b$ to $a^* \vee b^*$; the greatest-lower-bound property follows. The absorption laws of Proposition 3.4(4) then exhibit $(X, \vee, \wedge)$ as a lattice, bounded by (2). Distributivity of $\vee$ over $\wedge$ is not asserted and indeed fails in general. [AI-check: If you want a concrete non-distributive example here, add the diamond M_3 with a suitable involution and check (TJ1)–(TJ2) for a chosen tensor; otherwise the claim “fails in general” should be softened or given a citation.] $\square$

The structure obtained from the join module together with (P) and (PJ) alone—a bounded, possibly non-distributive lattice with an antitone involution satisfying the De Morgan laws—is what we call a *Polar Lattice*; it coincides with the classical notion of a bounded i-lattice [5]. We return to this identification in the orientation table of Section 4.

3.4 The Fundamental Structure Theorem

Assembling the three layers yields the main theorem of this section.

Theorem 3.6 (Fundamental Structure Theorem). *Let $(X, \vee, \otimes, *, e_\vee, e_\otimes)$ be a Polar Semiring. Then:*

1. the dual structure $(X, \wedge, \oplus, *, e_\wedge, e_\oplus)$ is again a Polar Semiring;
2. the polarity $*$: $X \rightarrow X$ is an isomorphism from $(X, \vee, \otimes, *, e_\vee, e_\otimes)$ onto its dual;
3. the join order of the original structure and the meet order induced by the dual structure coincide, in the sense that $\leq = \leq^*$.

Proof. (1) We verify the eleven axioms for $(X, \wedge, \oplus, *, e_\wedge, e_\oplus)$. Axioms J0–J3 for $\wedge$ and T1–T3 for $\oplus$ are Proposition 3.3. Axiom P is unchanged. For the dual of (TJ1), using (4)–(7) and (TJ1) in the original structure,

$$a \oplus (b \wedge c) = (a^* \otimes (b^* \vee c^*))^* = ((a^* \otimes b^*) \vee (a^* \otimes c^*))^* = (a \oplus b) \wedge (a \oplus c).$$

For the dual of (TJ2), $a \oplus e_\wedge = (a^* \otimes e_\vee)^* = e_\vee^* = e_\wedge$ by (8) and (TJ2). Finally, the dual of (PJ) asserts $a \wedge b = b \iff b^* \wedge a^* = a^*$; by Proposition 3.4(2) and Proposition 3.5(1), both

sides translate to statements about $\leq$, namely $b \leq a$ on the left and $a^* \leq b^*$ on the right, and these are equivalent by (PJ).

(2) The map $*$ is a bijection by (P), and (6)–(8) say exactly that it sends $\vee$ to $\wedge$, $\otimes$ to $\oplus$, $e_\vee$ to $e_\wedge$, and $e_\otimes$ to $e_\oplus$; it commutes with itself trivially. Hence it is an isomorphism onto the dual structure.

(3) This is Proposition 3.5(1). $\square$

Remark 3.7. The theorem justifies the architectural reading of the axiom system: the algebraic layer supplies the raw modules, the reflection layer produces the dual modules for free from (P), and the single integration axiom (PJ) welds the two orders together, at which point the duality becomes a self-duality of the whole structure. In particular, applying the construction twice returns the original Polar Semiring, since $(a^*)^* = a$.

4 Positioning

This section answers a single question: where do Polar Semirings sit relative to bounded Girard algebras? Bounded Girard algebras are the comparison of interest because they are the standard residuated setting in which polarity is a derived operation [1, 3, 8], and because the comparison produces two theorems: every bounded Girard algebra induces a Polar Semiring (Section 4.2), while the converse fails (Section 4.3). For orientation only, the following table locates a few neighbouring classes; none of them is used in any proof below. Its first three rows concern the order-theoretic reduct $(X, \vee, \wedge, *)$ only.

Class	Relation to PS	Axiomatic difference
Polar Lattice (= bounded i-lattice [5])	order reduct of every PS	none (reduct)
De Morgan algebra [7]	special case of the order reduct	adds $\vee/\wedge$ distributivity
Boolean algebra	special case of De Morgan algebras	adds excluded middle
Semiring with involution [4]	PS minus (PJ)	drops order compatibility
Bounded Girard algebra	proper subclass of PS (Corollary 4.5)	adds a residual

4.1 Bounded Girard algebras

We recall the definition of a bounded Girard algebra, rewritten entirely in the notation of this paper. The residual $\rightarrow$ is the one primitive of the definition that has no counterpart in a Polar Semiring; it is the only symbol foreign to our signature, and it appears in this section only.

Definition 4.1 (Bounded Girard algebra). A *bounded Girard algebra* (BGA) is a structure

$$(X, \vee, \wedge, \otimes, \rightarrow, e_\otimes, e_\oplus, e_\wedge)$$

satisfying:

$$(X, \vee, \wedge) \text{ is a lattice,} \tag{BGA1}$$

$$(X, \otimes, e_\otimes) \text{ is a commutative monoid,} \tag{BGA2}$$

$$a \otimes b \leq c \iff b \leq a \rightarrow c, \tag{BGA3}$$

$$(a^*)^* = a, \quad \text{where } a^* := a \rightarrow e_\oplus, \tag{BGA4}$$

$$a \leq e_\wedge \text{ for all } a, \tag{BGA5}$$

where $\leq$ denotes the lattice order, $a \leq b : \iff a \vee b = b$. The remaining operations of the PS signature are derived: $e_\vee := e_\wedge^*$ and $a \oplus b := (a^* \otimes b^*)^*$.

[AI-check: Definition 4.1 follows the axiomatization recorded in the project notes as sourced from [1]; verify against the published formulation (in particular whether boundedness is stated via (BGA5) or via a bottom element) before submission.]

By (BGA3)–(BGA4), the polarity of a BGA is a derived operation, manufactured from the residual and the distinguished element $e_\oplus$. The question of this section is whether that manufacturing step is essential.

4.2 Every bounded Girard algebra induces a Polar Semiring

We first isolate the consequences of residuation that will be needed; all of them use only (BGA1)–(BGA3).

Lemma 4.2. *In any structure satisfying (BGA1)–(BGA3):*

1. $a \leq b \iff e_\otimes \leq a \rightarrow b$;
2. $b \otimes (b \rightarrow c) \leq c$;
3. $a \leq b$ implies $a \otimes c \leq b \otimes c$;
4. $a \leq b$ implies $b \rightarrow z \leq a \rightarrow z$; in particular, with $z := e_\oplus$, the map $*$ is antitone.

Proof. (1) Put $b := e_\otimes$ in (BGA3) and use $a \otimes e_\otimes = a$. (2) By (BGA3), $b \otimes (b \rightarrow c) \leq c$ is equivalent to $b \rightarrow c \leq b \rightarrow c$. (3) By commutativity, $b \leq c \rightarrow (b \otimes c)$ holds unconditionally (it is (BGA3) applied to $c \otimes b \leq b \otimes c$); from $a \leq b$ and transitivity, $a \leq c \rightarrow (b \otimes c)$, and (BGA3) converts this back to $a \otimes c \leq b \otimes c$. (4) By (3), $a \otimes (b \rightarrow z) \leq b \otimes (b \rightarrow z)$, and by (2) the right-hand side is $\leq z$; then (BGA3) gives $b \rightarrow z \leq a \rightarrow z$. $\square$

Theorem 4.3 (BGA $\Rightarrow$ PS). *Let $(X, \vee, \wedge, \otimes, \rightarrow, e_\otimes, e_\oplus, e_\wedge)$ be a bounded Girard algebra, and let $*$, $e_\vee$ be the derived operations of Definition 4.1. Then $(X, \vee, \otimes, *, e_\vee, e_\otimes)$ is a Polar Semiring, and its derived meet and order coincide with the lattice meet and order of the BGA.*

Proof. Table 1 lists, for each of the eleven PS axioms, the BGA facts from which it follows. Seven rows are immediate: (J0), (J1), (J2) are contained in the lattice axiom (BGA1); (T1), (T2), (T3) are (BGA2); and (P) is (BGA4). The four remaining rows are proved after the table.

Table 1: Derivation of the eleven PS axioms from the BGA axioms.

PS axiom	Statement	Derived from	Status
(J0)	$a \vee a = a$	(BGA1)	immediate
(J1)	associativity of $\vee$	(BGA1)	immediate
(J2)	commutativity of $\vee$	(BGA1)	immediate
(J3)	$a \vee e_\vee = a$	(BGA4), (BGA5), Lemma 4.2(4)	proof below
(T1)	associativity of $\otimes$	(BGA2)	immediate
(T2)	commutativity of $\otimes$	(BGA2)	immediate
(T3)	$a \otimes e_\otimes = a$	(BGA2)	immediate
(TJ1)	$\otimes$ distributes over $\vee$	(BGA3), Lemma 4.2(1,3)	proof below
(TJ2)	$a \otimes e_\vee = e_\vee$	(BGA3), (J3)	proof below
(P)	$(a^*)^* = a$	(BGA4)	immediate
(PJ)	polarity	(BGA4), Lemma 4.2(4)	proof below

(J3). By (BGA5), $a \leq e_\wedge$ for all a ; since $*$ is antitone (Lemma 4.2(4)), $e_\vee = e_\wedge^* \leq a^*$ for all a . As $*$ is surjective by (BGA4), every element of X is of the form a^* , so $e_\vee$ is the least element of the lattice, and $a \vee e_\vee = a$ follows.

(TJ1). The inequality $(a \otimes b) \vee (a \otimes c) \leq a \otimes (b \vee c)$ follows from Lemma 4.2(3) applied to $b, c \leq b \vee c$. Conversely, put $u := (a \otimes b) \vee (a \otimes c)$. From $a \otimes b \leq u$ and $a \otimes c \leq u$, axiom (BGA3) yields $b \leq a \rightarrow u$ and $c \leq a \rightarrow u$, hence $b \vee c \leq a \rightarrow u$, and (BGA3) converts this to $a \otimes (b \vee c) \leq u$.

(TJ2). Since $e_\vee$ is the least element, $e_\vee \leq a \otimes e_\vee$; and $a \otimes e_\vee \leq e_\vee$ is, by (BGA3), equivalent to $e_\vee \leq a \rightarrow e_\vee$, which holds because $e_\vee$ is least. Antisymmetry concludes.

(PJ). Unfolding (2), the axiom asserts $a \leq b \iff b^* \leq a^*$. The forward direction is Lemma 4.2(4). For the converse, apply Lemma 4.2(4) to $b^* \leq a^*$ to obtain $(a^*)^* \leq (b^*)^*$, and use (BGA4).

Finally, the derived meet of the induced PS coincides with the lattice meet: for the antitone bijection $*$, the element $(a^* \vee b^*)^*$ is the greatest lower bound of $\{a, b\}$ in $(X, \leq)$, which in the lattice (BGA1) is $a \wedge b$. $\square$

4.3 The impossibility theorem

We now show that Theorem 4.3 cannot be reversed: the quadratic model of Example 2.6 is a Polar Semiring that admits no bounded Girard algebra structure. In a BGA, the polarity is by definition $a^* = a \rightarrow e_\oplus$ for the residual $\rightarrow$ of $\otimes$; on $([0, 1], \leq)$ with $\otimes$ the ordinary product, the residual, when it exists, is uniquely determined by (BGA3):

$$a \rightarrow c = \max\{b \in [0, 1] : ab \leq c\} = \min(1, c/a) \quad (a \in (0, 1]), \quad 0 \rightarrow c = 1. \quad (9)$$

Thus extendability of the quadratic model to a BGA amounts to the existence of a dualizing constant $q \in [0, 1]$ with $a^* = a \rightarrow q$ for all a , and the theorem rules this out.

Theorem 4.4 (Impossibility). *Let $([0, 1], \vee, \otimes, *, 0, 1)$ be the quadratic model of Example 2.6. There exists no $q \in [0, 1]$ such that*

$$a \otimes b \leq q \iff b \leq a^* \quad \text{for all } a, b \in [0, 1]. \quad (10)$$

Consequently, the quadratic model cannot be extended to a bounded Girard algebra: no choice of residual and dualizing element $e_\oplus$ makes its polarity (3) arise as $a \mapsto a \rightarrow e_\oplus$.

Proof. Fix $a \in (0, 1]$. As functions of b , both sides of (10) are threshold conditions: the left-hand side holds precisely for $b \leq \min(1, q/a)$, the right-hand side precisely for $b \leq a^*$. Since (10) must hold for every $b \in [0, 1]$, the thresholds must agree:

$$\min(1, q/a) = a^* \quad \text{for all } a \in (0, 1]. \quad (11)$$

Evaluating (11) at $a = 1$ gives $\min(1, q) = 1^* = 0$; as $q \in [0, 1]$, this forces $q = 0$. But substituting $q = 0$ into (11) at any $a \in (0, 1)$ gives $0 = a^* = (1 - \sqrt{a})^2 > 0$, a contradiction. Hence no q satisfies (10).

For the final claim: in any BGA structure on this carrier extending the given $(\vee, \otimes)$, the order is the usual order of $[0, 1]$, the residual is forced to be (9) by (BGA3), and the polarity is $a \rightarrow e_\oplus$ for the primitive constant $q := e_\oplus$; the displayed equivalence (10) is then exactly the requirement that this derived polarity coincide with (3). [AI-check: Per final-rules §6.1, step (11) (the “(†)” step of the source notes) required independent verification. I have re-derived it above and it checks out: for fixed $a \in (0, 1]$ the LHS of (10) is $b \leq \min(1, q/a)$, the RHS is $b \leq a^*$, equality of the two down-sets on $[0, 1]$ forces equality of thresholds; $a = 1$ forces $q = 0$; $q = 0$ fails for every $a < 1$. Please perform your own final check (and, if desired, a Mace4 confirmation on a finite subchain) before submission.] $\square$

Corollary 4.5. $\text{BGA} \subsetneq \text{PS}$: the class of Polar Semirings induced by bounded Girard algebras via Theorem 4.3 is a proper subclass of the class of all Polar Semirings.

Remark 4.6. The impossibility has a structural explanation. Among the eleven PS axioms, none constrains $\otimes$ and $*$ jointly through the order: (TJ1)–(TJ2) relate $\otimes$ to $\vee$ without mentioning $*$, while (P) and (PJ) relate $*$ to $\vee$ without mentioning $\otimes$. Whether a given polarity is residuation-compatible with a given tensor is therefore left entirely undetermined by the axioms, and Theorem 4.4 shows that this freedom is genuine: the gap between the two classes is non-empty. In this precise sense, Polar Semirings are a residuation-free generalization of bounded Girard algebras—the polarity is promoted to a primitive, the residual is discarded, and Theorems 4.3 and 4.4 together certify that the generalization is strict rather than a reformulation.

5 Conclusion

We have introduced Polar Semirings, an axiomatization of polarity as a primitive algebraic notion, and established their basic theory and their position in the landscape of involutive structures. Three results summarize the paper. First, the eleven axioms of Definition 2.1 are independent and admit natural models on chains, including Boolean, tropical, and quadratic models (Section 2). Second, the axioms decompose into an algebraic layer, a free reflection layer generated by the involution, and a one-axiom integration layer, and this architecture yields the Fundamental Structure Theorem: every Polar Semiring is canonically self-dual, with the polarity acting as an isomorphism onto the dual structure and the two induced orders coinciding (Theorem 3.6). Third, every bounded Girard algebra induces a Polar Semiring upon forgetting its residual (Theorem 4.3), whereas the quadratic model on the unit interval admits no bounded Girard algebra extension (Theorem 4.4); hence $\text{BGA} \subsetneq \text{PS}$ (Corollary 4.5), and the generalization is strict.

The conceptual message is that the dual architecture usually attributed to residuation—dual monoids, De Morgan laws, order reversal—is in fact generated by three compatibility axioms, (PJ), (TJ1), and (TJ2), which are jointly much weaker than the residuation law (BGA3).

Several directions remain open. On the axiomatic side, one may ask for the exact strength of residuation over the PS axioms: which additional axioms, formulated in the PS signature, characterize the Polar Semirings that do extend to bounded Girard algebras? On the model-theoretic side, the variety generated by Polar Semirings invites study of its free objects, its subdirectly irreducible members, and the decidability of its equational theory. On the structural side, the non-commutative case, in which (T2) is dropped and the polarity may split into left and right polarities, and the distributive case, in which the induced lattice is required to be distributive, both appear tractable with the methods of Section 3. [AI-check: If you have partial results on any of these directions (e.g. the characterization of residuable Polar Semirings), consider replacing the corresponding sentence with a precise conjecture.]

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